Medical Education Industry - AI Employees Register

The medical education industry, function by function - AI Employees register

Owner's ask (4 Nov 2026): *"NOW A MEDICAL EDUCATION INDUSTRY AI EMPLOYEE LIST"*. Sixth in the series after `healthcare_industry/` (160 roles), `pharmacy_industry/` (118), `hospital_industry/` (128), `medical_devices_industry/` (118) and `diagnostics_industry/` (108). This one is the industry that TRAINS the people the others employ: MBBS to super-speciality, nursing, dental, pharmacy, allied health, AYUSH, CME/CPD, entrance and licensing-exam preparation, simulation, the medical college as an institution, and the education-technology and publishing businesses around it. Roles already in earlier registers (HC#147-153, PH#117) are cross-referenced and not counted.

12 fields, 107 candidate AI Employees.

How honest this is

- No new web research for this file. It is **domain knowledge, not a sourced census**. Bodies and rules named are from memory and **must be re-checked on primary text** before building: NMC (Competency-Based Medical Education curriculum, logbooks, minimum standard requirements, attendance rules, NExT licensing exam plan), NBEMS (DNB), INC (nursing), DCI (dental), PCI (pharmacy), NCISM / NCH (AYUSH), NAAC / NIRF (institution rankings), state health-university rules, NEET-UG / NEET-PG / FMGE counselling (MCC), UGC rules for deemed universities, and abroad ACGME / USMLE / PLAB / GMC / AMC requirements. NExT and several NMC rules have changed repeatedly: check current status.
- Nothing is built. Hoola already has a strong head start: **Guru Sahayak** (personal tutor, hint ladder, mastery, SM-2 spaced repetition, exam mode, guardian card), **Shiksha** (classes, learners with consent, question bank, gradebook), **Pathyakram** (curriculum & exam designer), **Daftar** (school office), **Shodh** (research assistant), **pylab** (locked Python), **Prayogshala** (3D practical labs) - all in `server/app/` (guru, shiksha, pathyakram, daftar, shodh, prayog). A medical-education desk would extend them, not start from zero.

The findings that shape every medical-education desk

1. The licence belongs to the teacher, the examiner and the institution - never to the software

Same ladder as before: **A** admin/prepare, **B** assist (draft / flag / tutor for a named human), **C** reserved. Reserved here: summative marks and pass/fail decisions, entrustment ("this resident may do it unsupervised"), signing a logbook or a clinical competency assessment, admission/counselling allotment, faculty appointment and promotion, accreditation declarations, cheating/misconduct verdicts, proctoring decisions against a student, and medical advice to a real patient disguised as "practice".

2. Students are the product's end user, and the biggest honest win is TUTORING and PRACTICE

Medical students and residents drown in volume: anatomy, pharmacology, case reasoning, drug names, exam banks. A tutor that explains, quizzes, spaces review (SM-2), builds image-based questions, runs a **virtual patient** that the student interviews, and shows weak areas is lawful, safe and sticky. Rules: cite the textbook/guideline, show the hint ladder, mark

everything "for study, not for patient care", never hand a real patient's plan. Guru's exam mode (hints only) already exists for supervised settings.

3. Competency-based education is a paperwork engine - tier A at scale

CBME demands logbooks, certifiable skills, SDL/AETCOM entries, attendance (75/80% rules), internal-assessment records, rotation schedules, research/thesis milestones and faculty development hours, all signed by someone. Tracking, reminders, completeness checks and report assembly are exactly where staff time disappears, with no clinical risk.

4. Assessment integrity and bias are the regulated edge

AI for grading, proctoring, admissions or "readiness" scoring is high-risk in the EU AI Act and sensitive everywhere: bias, appeals, transparency, data protection (DPDP; students may be minors at 17). Any tier-B role that scores a human needs a documented rubric, a second human on high-stakes cases, and an appeal path.

THE REGISTER (AI Employee - what it does - tier - the human who decides)

1. Pre-admission: entrance coaching & counselling

1. NEET / Entrance Study Planner - syllabus map, weekly plan, spaced repetition, weakness-driven revision (Guru Sahayak) - A - student / parent
2. Entrance MCQ Coach - adaptive practice with worked solutions and error taxonomy; no leaked papers, no fake "predicted" paper claims - A - coaching faculty
3. Mock-test Runner & Rank Estimator (percentile from past public cut-off data, pylab; **estimate labelled as estimate**) - A - coaching faculty
4. Counselling Choice-list Assistant (MCC / state counselling rules, past closing ranks, category & domicile rules; **student decides**) - A - counsellor
5. College Compare Desk (fees, seats, bond/stipend, clinical material, NMC recognition status) - A - counsellor
6. Document Checklist & Form Filler (certificates, domicile, income, reservation, medical fitness) - A - parent / counsellor
7. Scholarship & Education-loan Finder (state schemes, banks, repayment calculators) - A - family
8. Study-abroad Pathway Advisor (FMGE route, MBBS abroad country vs NMC rules, recognition warnings) - B - licensed overseas-education counsellor
9. Parent Support Agent (stress, expectations, FAQs; refers mental distress to a human helpline) - A/B - counsellor

2. Undergraduate medical education (MBBS) - tutoring & learning

10. Anatomy Tutor (dissection-room prep, structure quizzes, radiology cross-sections, 3D via Hoola engines) - A - anatomy faculty
11. Physiology & Biochemistry Concept Tutor (pathway walk-throughs, numeric problems in pylab) - A - faculty
12. Pharmacology Mnemonic & Drug-class Coach (class tables, ADR/interaction recall) - A - faculty
13. Pathology & Microbiology Image Quiz Builder (slide / gram-stain / culture sets with explanations) - A - faculty
14. Clinical Reasoning Coach (case vignette -> differentials -> investigations -> management reasoning, Socratic) - B - clinical faculty
15. Virtual Patient Simulator (history-taking roleplay with scripted case bank, finds missed questions; **study only**) - B - clinical faculty
16. OSCE / Viva Rehearsal Partner (station scripts, timer, checklist feedback) - A/B - skill-lab faculty

17. Practical Lab Companion (Prayogshala-style virtual experiments, results to journal) - A - faculty
18. Community-medicine / Biostatistics Helper (sample size, epidemiology exercises, survey design in pylab) - A - PSM faculty
19. Forensic Medicine & Toxicology Case-law Reader (statutes, case summaries with sources) - A - faculty
20. Language Support Tutor (Hinglish / regional explanations of the same concept; English-medium exam terms kept) - A - faculty
21. Peer Study Group Organiser (rooms, spaced topic rotation, doubt-queue to faculty) - A - class coordinator
22. Student Wellbeing Check-in (burnout scales, sleep/habits nudges; **human counsellor hand-off, crisis protocol**) - B - college counsellor

3. Postgraduate & super-speciality training (MD/MS/DNB/DM/MCh)

23. NEET-PG / NExT-style Exam Coach (subject maps, error log, timed practice) - A - student
24. Thesis & Dissertation Coach (research question, design, sample size, ethics, statistics shell, reference formatting; student's own data and writing) - B - guide / supervisor
25. Literature Review Assistant (search strategy, screening log, extraction tables, citation check via Shodh) - B - guide
26. Case Presentation & Journal-club Preparer (structured slides outline, critical-appraisal checklists CONSORT / STROBE / PRISMA) - A - resident
27. Procedure Log Keeper (counts, supervision level entered by supervisor, gaps vs required numbers) - A - HOD
28. Rotation & Duty Roster Planner (work-hour limits, postings, leave swaps) - A - PG coordinator
29. Speciality Board Exam Prep (clinical + viva; mirrors NBEMS / international board styles) - A - resident
30. Research-methods Statistician Assistant (pylab analysis templates, assumption checks, plots; **interpretation reviewed by statistician**) - B - biostatistician
31. Manuscript Preparation Assistant (IMRaD scaffolds, guideline checklists, journal-fit lookup; no fabricated data/refs) - B - authors

4. Nursing, dental, pharmacy & allied-health education

32. Nursing Student Tutor (fundamentals, care plans, NCLEX-style and INC-syllabus practice) - A - nursing faculty
33. Nursing Clinical-skills Checklist Coach (injection, catheter, wound care step checklists; competency signed by clinical instructor) - A/B - clinical instructor
34. Nursing Care-plan Practice Generator (NANDA/NIC/NOC patterns on case scenarios) - A - faculty
35. Dental Education Assistant (tooth charting practice, radiograph quiz, case-based planning exercises) - A - dental faculty
36. Pharmacy Student Companion (D.Pharm / B.Pharm / Pharm.D: PH#117 plus dispensing role-play) - A - faculty
37. Physiotherapy / OT / MLT / Radiography Skills Tutors (technique checklists, case drills) - A - respective faculty
38. Paramedic & EMT Scenario Trainer (protocol drills, triage practice with synthetic scenes) - A/B - EMS instructor
39. Community Health Worker / ASHA Training Companion (voice-first, local language, job-aid lessons and refreshers) - A - district health trainer

5. AYUSH & traditional medicine education

40. Ayurveda Student Tutor (Sanskrit shloka explanation with source, dravyaguna tables, practical clinical correlation) - A - BAMS faculty
41. Unani / Siddha / Homeopathy Study Companions (materia medica recall, repertory exercises; source-cited) - A - respective faculty
42. Yoga & Naturopathy Course Assistant (asana libraries, anatomy cross-links, safe-practice cautions) - A - faculty

43. Integrative Medicine Evidence Reader (RCT summaries with grade and limits) - B - research faculty

6. Continuing education (CME / CPD) & faculty development

44. CME Credit Tracker (state / national council credits, certificates, renewal windows; HC#150) - A - doctor / council
45. Personalised CME Recommender (gap-based from the clinician's practice mix and recent guideline changes) - A - doctor
46. Guideline Update Digest (new ICMR / WHO / specialty-society guidelines summarised with source and date) - B - clinician reviews
47. Journal Club Moderator for Practitioners (weekly paper, critical questions, quiz) - A - society
48. Case-based Quiz for Practising Clinicians (anonymised cases, rationale, references) - A/B - CME director
49. Faculty Development Programme Planner (MET, assessment writing, research methods; hours tracked) - A - medical-education unit
50. Educator Teaching-portfolio Builder (evidence, feedback summaries, promotion file shell; **promotion by committee**) - A - faculty

7. Assessment: questions, exams & grading

51. MCQ / SAQ / LAQ Item Writer (blueprint-aligned, Bloom level tags, key and distractor rationale; **faculty reviews every item**) - B - subject faculty (extends Pathyakram)
52. Item Analysis & Question-bank Curator (difficulty, discrimination, distractor efficiency, retire / revise list via pylab) - A - assessment cell
53. Exam Blueprint & Paper Assembler (CBME competency map, mark weight, no repeat across sittings) - A - exam controller
54. OSCE Station Designer & Examiner Pack Builder (checklists, global-rating anchors, standardised-patient scripts) - B - OSCE coordinator
55. Rubric-based Draft Grader for Written Answers (marks per rubric criterion with quoted evidence; **second human marks all high-stakes**) - B - examiner decides
56. Practical / Viva Scheduling Coordinator (examiner pairing, conflict-of-interest checks, room logistics) - A - exam cell
57. Internal Assessment Tracker (continuous assessment, eligibility for university exam, attendance thresholds) - A - IQAC / registrar
58. Result Processing & Moderation Assistant (consolidation, grace-mark rule application, tabulation checks; **declaration by controller**) - A - controller of examinations
59. Re-evaluation & Grievance Handler (request intake, anonymised re-mark packet) - A - exam cell
60. Plagiarism & Authorship-integrity Screener (similarity reports, reference-existence checks; **verdict by committee**) - B - research integrity officer
61. Proctoring Flag Reviewer (online exams: flags only, no automatic penalty, appeal path) - B - exam controller

8. Clinical training operations: wards, simulation & skills labs

62. Simulation-lab Scheduler & Equipment Manager (mannequins, task trainers, consumables, calibration) - A - simulation director
63. Scenario Designer (debrief scripts, learning objectives, observer checklists for emergency / OB / paediatric / OT scenarios) - B - simulation faculty
64. Debrief Note Assistant (transcript to structured feedback by objective; **facilitator-led**) - B - facilitator
65. Standardised-patient Trainer & Scheduler - A - skills-lab coordinator

66. Clinical Posting Allocator (students x wards x batches, fairness, leave) - A - clinical coordinator
67. Competency Sign-off Tracker (skills required, observed attempts, assessor name, expiry; **sign-off by assessor only**) - A - clinical coordinator
68. Entrustable Professional Activity (EPA) Dashboard (levels recorded by supervisors; progression recommendation to the committee) - B - competency committee
69. Logbook Completeness Checker (missing signatures, wrong formats, inconsistent counts) - A - HOD
70. Patient Safety Learning Packager (de-identified incidents turned into teaching cases) - B - patient-safety + education lead
71. Rural Posting / Internship Coordinator (placements, stipends, housing, travel via Maps) - A - internship coordinator

9. Institution management: the medical college as an organisation

72. Admissions Office Assistant (post-counselling document verification, fee collection, hostel allotment) - A - registrar
73. Student Records & Attendance Tracker (biometric / app attendance, shortage notices, condonation files) - A - registrar
74. Fee, Stipend & Scholarship Accountant (instalments, bond recovery, stipend sheets, TDS) - A - accounts
75. Hostel, Mess & Transport Manager - A - warden / administrator
76. Library & Resource Desk (e-resources access, citation help, interlibrary requests, subscription analytics) - A - librarian
77. NMC / University Compliance Calendar (annual disclosure, minimum standard requirements, faculty-student ratios, inspection prep; **declarations by dean**) - A - dean / compliance cell
78. Inspection-readiness Auditor (checklists vs MSR, evidence binder by department, gap list) - B - dean
79. NAAC / NIRF Data Compiler (criteria mapping, evidence indexing, narrative drafts; **submission by IQAC head**) - A/B - IQAC head
80. Faculty Records & Eligibility Checker (qualification, experience-years, publications requirement for designations) - A - registrar
81. Faculty Recruitment & Roster Assistant (advertisements, shortlisting checklist; **panel decides**) - A - HR / registrar
82. Student Grievance & Anti-ragging Cell Assistant (case intake, timeline, mandatory-report prompts; **committee acts**) - B - anti-ragging committee
83. Mental Health & Student Support Cell Assistant (appointment triage to human counsellors, anonymised trend reporting) - B - counsellor
84. Alumni & Placement Desk (internship / residency opportunities, fellowships, alumni mentoring match) - A - alumni cell
85. College Finance & Budget Controller (department budgets, equipment purchases, funding gaps, pylab forecasts) - A - finance officer
86. Research Cell Manager (ethics-committee scheduling, funding calls, ICMR / DST / DBT grant trackers) - A - research cell head
87. Institutional Ethics Committee Secretariat (agenda, protocol packets, minutes, SAE flow; HC#153) - A - member secretary

10. Licensing, credentials & international movement

88. Medical Registration Helper (state council registration, NMC portal documents, renewal reminders) - A - doctor
89. Foreign-graduate Licensing Pathway Planner (FMGE / NExT, internship equivalence, document order) - A - student

90. USMLE / PLAB / AMC / MCCQE Prep Coach (step-wise plan, question practice, error log) - A - student
91. Residency Application Assistant (CV, personal statement structure, LOR tracker, programme research, interview practice; **applicant's own words**) - A/B - applicant / mentor
92. Credential Verification Packager (primary-source verification requests ECFMG / state, notarisation checklist) - A - applicant
93. Visa & Relocation Planner for Clinicians - A - applicant
94. CPD Revalidation File Builder (UK GMC appraisal, other jurisdictions: reflective logs, feedback, evidence) - A - doctor

11. Publishing, ed-tech & content businesses

95. Medical Textbook Content Editor Assistant (consistency checks, reference verification, figure-credit tracking; **author / editor approve**) - B - editor
96. Question-bank Product Manager (coverage maps, user-error analytics, update cycles) - A - product lead
97. Lecture Video Producer (slides to narrated animated video; Hoola FRAMES; accuracy review gate) - B - faculty reviewer
98. Mnemonic & Visual-Aid Designer (concept diagrams, flashcards, image-occlusion cards) - A - faculty
99. Translation & Localisation Editor (Hindi / regional medical terms, glossary control) - B - bilingual faculty
100. Learner Analytics Analyst (cohort retention, weak-topic heatmaps, churn, pylab) - A - ed-tech lead
101. Accessibility Assistant (captions, alt-text, screen-reader-safe slides, dyslexia-friendly formats) - A - content lead
102. Integrity & Copyright Checker for Course Content (licence trail for images, no scraped paywalled material) - B - legal / editor
103. Community Forum Moderator (answers FAQs, flags unsafe advice or exam-leak posts; **removals by moderator**) - B - community manager

12. Public engagement & patient/health education (the lay audience)

104. Patient Education Material Writer (reading-level checked, multi-language, source-cited, clinician-approved; HC#108) - B - clinician approves
105. School Health Education Planner (hygiene, menstrual health, vaccination awareness sessions) - A - school health officer
106. Health-literacy Quiz & Myth-buster (WHO / MoHFW-sourced) - A - public-health educator
107. Community Training for First Aid & CPR (certified instructor curriculum, scheduling, certificate tracking) - A - certified instructor

NEVER an AI Employee (tier C)

Declare marks, results, pass/fail or ranks; sign a logbook, competency or entrustment decision; allot a seat or counselling choice; certify attendance condonation; hire, promote or discipline faculty; rule on misconduct, cheating, ragging or plagiarism; submit accreditation or inspection declarations; give a real patient a plan under the label "study"; impersonate a supervising doctor; use a student's data for model training without consent.

Suggested build order

1. **Wave 1 - extend what Hoola already has (tier A, student-facing)**: 1, 2, 10-13, 16-18, 20-21, 23, 32, 44, 88, 90 on Guru Sahayak + Prayogshala + SM-2.
2. **Wave 2 - college paperwork engine (tier A/B)**: 27, 28, 57, 67, 69, 73, 77-80, 85-87.

3. **Wave 3 - assessment and simulated-patient tools (B, human-signed)**: 14, 15, 24, 51, 52, 54, 55, 58, 61, 63, 68. Fit to Hoola: department **"Medical Education"** beside Healthcare / Pharmacy / Devices / Diagnostics; reuse Guru Sahayak (hint ladder, mastery, SM-2, exam mode), Shiksha (classes, question bank, gradebook), Pathyakram (blueprints, rubrics), Daftar (office), Shodh (research), pylab (statistics), Prayogshala (virtual labs), HOOLA FRAMES (lecture videos), approval cards. Best first pilot: one NEET-PG / NEET-UG coaching institute or a single nursing college - tutoring and practice need no clinical sign-off, and a coaching business buys on results within a season.

Open questions for the owner

- Which learner first: **NEET-UG aspirants**, **MBBS students**, **PG/NEET-PG residents**, **nursing students**, or **practising doctors (CME)**? Each has a different buyer, a different price, and a different exam calendar.
- Sell to the student (B2C, low ticket, many) or to the institution (B2B, one decision-maker, deployment through the college)? The second needs the Wave 2 paperwork employees and sets the data-protection bar higher.
- Which language mix leads (English, Hinglish, Hindi, Marathi...)? Medical exams are English, but understanding often starts in the mother tongue; the register assumes bilingual explanation with English exam terms intact.
- Will Hoola use real exam questions/books? Copyright and leaked-paper risk are serious; the register assumes faculty-authored or properly licensed content only.
- "GO per field" turns each into a sourced report with NMC / university rules checked on primary text.